

Gate-Driver Selection and Gate-Loop Layout for Paralleled GaN: Current Sharing, Switching Loss, Negative Off-Voltage and Scaling

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Summary.

- **Layout decides the sharing.** With equal gate loops and Kelvin sources two paralleled transistors share current and switching energy exactly; a gate-loop mismatch of 2 versus 8 nH gives 19 % dynamic imbalance and a 4:1 split of E_{on} , and a common-source mismatch without Kelvin 29 % and 2.8 times E_{on} in one device. The centre-driver star gives 1.71 nH per transistor in both cells and a common-source inductance of 0.030 nH (Q3D), against 0.061 nH without Kelvin.
- **Sink strength sets the turn-off, not the gate bump.** Raising the driver pull-down from 0.05 to 0.5 Ω increases E_{off} from 7.6 to 10.5 μJ but the high-side bump only from 1.07 to 1.12 V: the bump is the Miller charge divided onto C_{GS} (≈ 1.06 V), faster than any hold-off path can react.
- **Negative off-voltage removes the bump at a small dead-time cost.** With -1 V the peak high-side gate voltage falls from 0.99 to -0.01 V, the undershoot stays at -2.3 V (rating -4 V), and the reverse-conduction drop during the dead time rises by 1 V: dead-time loss $0.20 \rightarrow 0.29$ W per leg at 10 ns and 50 kHz (0.38 W at -2 V). Net leg efficiency changes by about 0.01 %. -1 V is selected; -2 V leaves only 0.7 V to the -4 V limit.
- **Driver: 2EDF7275K**, dual-channel isolated, 0.35 $\Omega/8$ A sink, 4 V UVLO, so that a split $+5/-1$ V supply per channel is possible. Its peak sink current for two transistors at -1 V is about 10.9 A, above the 8 A rating; an external turn-off resistor of about 0.4 Ω per transistor, or the driver's own current limit, keeps it in range.
- **More transistors need more drivers.** One channel can drive two transistors symmetrically. Sharing it among 3, 4 or 6 raises the switching energy per transistor by 16, 31 and 60 % (output resistance per transistor scales with n), raises the peak sink current to 12–14 A, and a one-driver tree adds gate-loop inductance, which brings back part of the bump. Paralleling therefore scales in pairs, one driver channel per pair.

1 Gate-drive circuit of the building block

Each switch consists of two EPC2361 in two mirrored commutation cells. One driver sits on the symmetry line and drives both transistors of a switch through a star with Kelvin source returns (Fig. 1). Per transistor the gate path consists of the driver pull-up or pull-down, the external $R_{\text{g,on}} = 2 \Omega$ (with a diode for turn-off) or $R_{\text{g,off}} = 0 \Omega$, the star branch L_G and the internal 0.4 Ω gate resistance. The driver output stage is shared by the two transistors, so its resistance counts twice per transistor. All simulations use the LTspice building-block model with the Q3D power loop and gate star at 75 V and 141 A.

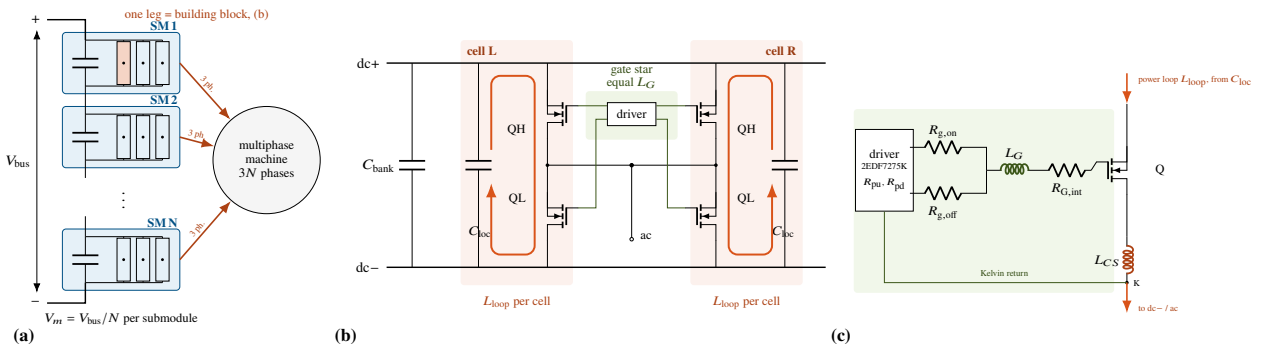


Figure 1: (a) SPB converter, (b) building block with the centre driver and two mirrored cells, (c) gate and power circuit of one transistor with the Kelvin connection K, which limits the common-source inductance L_{CS} to the package and pad.

2 Layout: current distribution and switching loss

Paralleled GaN transistors share the conduction current well because $R_{DS(on)}$ has a positive temperature coefficient; the static current at the end of a pulse is 59–60 A per transistor in every case of Fig. 2. The switching transients, however, are set by the gate circuit of each transistor, and any asymmetry moves switching energy into one device (Fig. 2, Table 1).

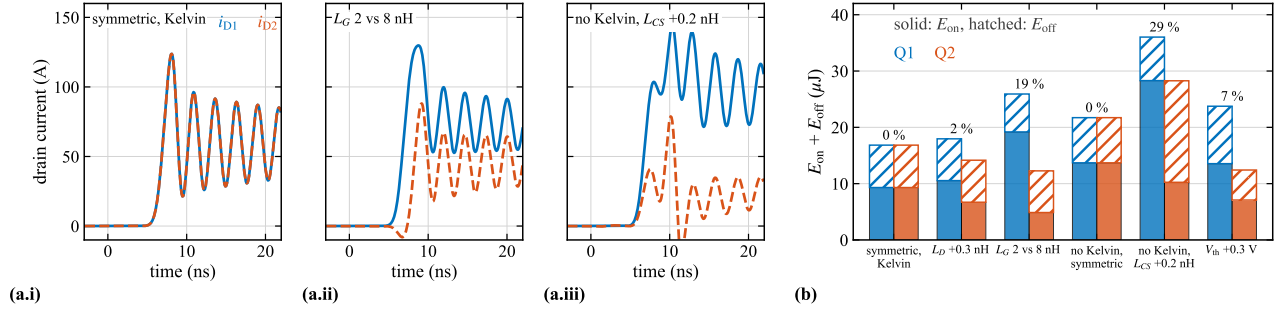


Figure 2: Two paralleled EPC2361 (LTspice, 75 V, 120 A): (a) drain currents at turn-on for the symmetric layout, a gate-loop mismatch and a common-source mismatch without Kelvin; (b) turn-on (solid) and turn-off (hatched) energy of both transistors for one asymmetry at a time, with the dynamic imbalance.

Table 1: Sensitivity of sharing and switching energy to one asymmetry at a time

Case	I_{diff} [%]	E_{on} Q1/Q2 [μJ]	E_{off} Q1/Q2 [μJ]	mechanism
symmetric, Kelvin	0	9.3 / 9.3	7.5 / 7.5	—
drain path +0.3 nH	2	10.5 / 6.7	7.4 / 7.5	slower di/dt in one branch
gate loop 2 vs 8 nH	19	19.2 / 4.9	6.7 / 7.4	one gate delayed by $\approx \Delta L_G / R$
no Kelvin, symmetric	0	13.7 / 13.7	8.0 / 8.0	negative feedback, both slower
no Kelvin, $L_{CS} + 0.2$ nH	29	28.3 / 10.2	7.7 / 18.0	feedback stronger in one device
$V_{th} + 0.3$ V	7	13.6 / 7.1	10.2 / 5.3	device spread, not layout

Three layout rules follow, in order of importance:

- 1. Kelvin source for every transistor.** A common-source inductance acts as negative feedback on the gate; it slows the switching when it is equal, and it unbalances it when it is not. The Q3D extraction of the combined power and gate model gives 0.030 nH with a Kelvin via in each source pad and 0.061 nH when the driver ground is taken from the dc-plane; even 0.030 nH induces about 0.8 V at 27 A/ns, but equally in both transistors.
- 2. Equal gate loops.** A mismatch ΔL_G delays one gate by roughly $\Delta L_G / R_G$, comparable to the 4 ns current rise. A side-mounted driver would give 2.0 and 7.7 nH to the near and far transistor (Q3D), i.e., close to the 2 versus 8 nH case; the centre driver gives 1.71 nH to both.
- 3. Power-loop symmetry** matters less: 0.3 nH in one drain path gives 2 % imbalance, mainly a shift of E_{on} . Threshold spread (7 %) is a device property and can only be reduced by selection.

3 Driver output stage: sink and source

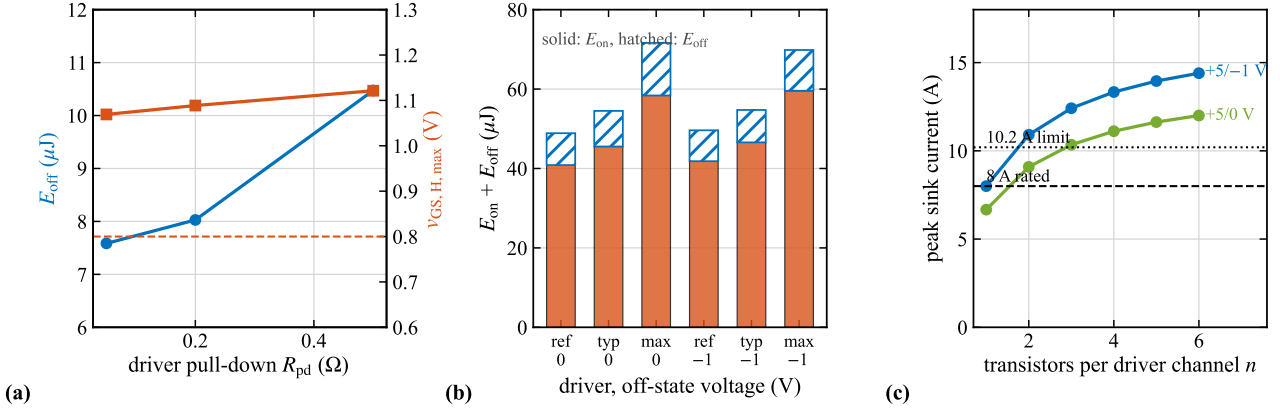


Figure 3: (a) Turn-off energy and high-side gate bump against the driver pull-down resistance (0 V off-state). (b) Switching energy for the LT8418-like reference (0.6/0.2 Ω), the 2EDF7275K with typical (0.85/0.35 Ω) and maximum (1.6/0.75 Ω) output resistance, at 0 and -1 V off-state. (c) Peak sink current of one driver channel against the number of transistors it drives (0.35 Ω pull-down, 0.4 Ω internal gate resistance, no external turn-off resistor), with the 2EDF7275K rating and current limit.

The **sink** (pull-down) path does two things: it turns the transistor off and it holds the complementary transistor off. For the turn-off, it is part of $R_{\text{g,off,tot}}$; as long as this stays below the critical resistance $R_{\text{crit}} = V_{\text{pl}}Q_{\text{oss,tot}}/(IQ_{\text{GD}})$ (about 1.5 Ω at 139 A), the channel is off before v_{DS} rises and E_{off} stays small (7–8 μJ); above it the turn-off becomes channel-controlled and E_{off} grows quickly (18.5, 39.6 and 78.7 μJ for 2, 5 and 10 Ω external, slow-switching study). Fig. 3a shows the same trend for the driver itself: 7.6, 8.0 and 10.5 μJ for 0.05, 0.2 and 0.5 Ω . For the hold-off, in contrast, even 0.05 Ω does not help (1.07 V bump against 1.12 V at 0.5 Ω), because the Miller charge arrives within 4–6 ns and the gate loop ($L_{\text{G}} = 1.7$ nH, $R_{\text{G,int}} = 0.4$ Ω) cannot move it in that time: $\Delta v_{\text{GS}} \approx Q_{\text{GD}}/C_{\text{GS}} \approx 1.06$ V. A strong sink is therefore needed for a fast, low-loss turn-off and for holding against slower disturbances, but the fast bump has to be handled by the off-state voltage (Section 4).

The **source** (pull-up) sets, together with $R_{\text{g,on}}$, the turn-on speed and E_{on} . The 2EDF7275K has a weaker pull-up than the reference (0.85 against 0.6 Ω), which raises E_{on} by about 11 % (typical) and 43 % (maximum resistance, Fig. 3b); $R_{\text{g,on}}$ can be reduced from 2 to 1.5 Ω to compensate, as long as the high-side overshoot stays below 90 V.

The **peak sink current** is often the binding rating. With -1 V and no external turn-off resistor, one channel driving two transistors must sink about $6 \text{ V}/(0.35 + 0.4/2) \Omega \approx 10.9$ A (Fig. 3c), more than the 8 A rating of the 2EDF7275K; its active current limit (about 10 A) then shapes the turn-off. An external turn-off resistor of about 0.4 Ω per transistor limits the peak to 8 A at the cost of a slightly slower turn-off (per-transistor $R_{\text{g,off,tot}} \approx 1.5 \Omega \approx R_{\text{crit}}$ at the peak current); this has to be confirmed with a current-limited driver model.

4 Negative off-voltage

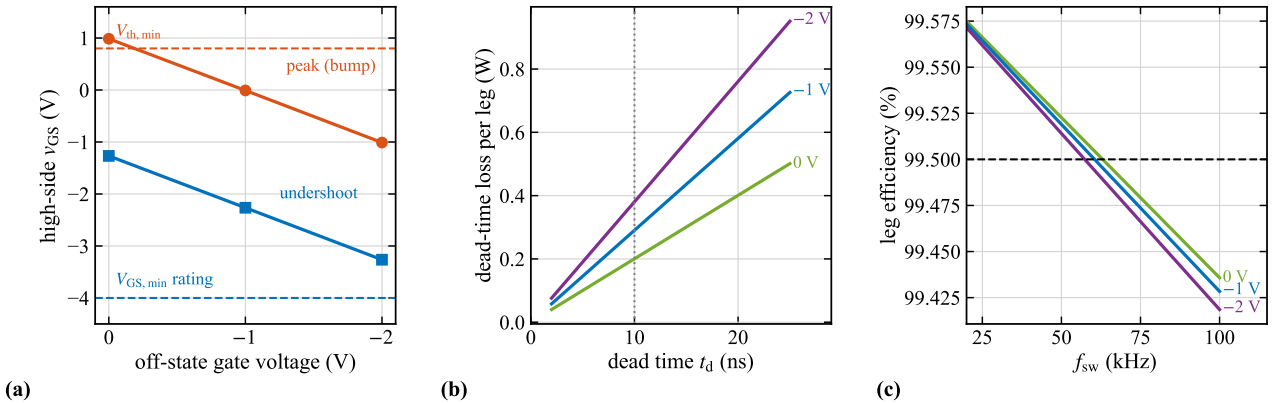


Figure 4: (a) Peak (bump) and minimum (undershoot) high-side gate voltage against the off-state voltage (75 V, 141 A). (b) Dead-time loss per leg against dead time for 0, -1 and -2 V (50 kHz, 100 A_{rms}). (c) Leg efficiency against switching frequency with the 2EDF7275K (typical) for the three off-state voltages, including the change of the switching energies and the dead-time loss.

A negative off-state voltage shifts the whole high-side gate waveform down by the same amount (Fig. 4a): the peak falls from 0.99 to -0.01 V at -1 V and to -1.01 V at -2 V, while the bump amplitude itself stays about 1 V. The price appears in three places:

- **Dead-time loss.** A GaN HEMT conducts in reverse through its channel with $v_{SD} \approx V_{th} + |v_{GS,off}| + iR$; every volt of negative bias adds one volt to the reverse drop during the dead time. At 10 ns and 50 kHz the dead-time loss per leg rises from 0.20 to 0.29 and 0.38 W (Fig. 4b); it grows linearly with the dead time (0.15 and 0.58 W at 5 and 20 ns with -1 V) and with f_{sw} , so a negative bias should be combined with a short, well-controlled dead time.
- **Undershoot.** When the complementary transistor turns off, the gate is pulled further negative by the Miller current: -1.3, -2.3 and -3.3 V for 0, -1 and -2 V, against an absolute maximum of -4 V; -2 V leaves only 0.7 V.
- **Switching energies.** The faster turn-off slightly raises the low-side overshoot (86.5 → 88.7 V at -1 V) and the larger gate swing (6 V) increases the gate charge; the switching energy changes by less than 2 %.

The net effect on the leg efficiency is about 0.01 % per volt at 50 kHz (Fig. 4c). -1 V gives 0.8 V margin to $V_{th,min}$ and is selected. The bias requires a driver whose output ground can be referenced below the transistor source, i.e., an isolated output with a split supply (2EDF7275K: +5 V/-1 V from a 6 V isolated rail and a Zener around the Kelvin source); 5 V-class bootstrap half-bridge drivers (maximum supply about 5.9 V) cannot provide it without losing on-state gate voltage.

5 More transistors in parallel: why more drivers

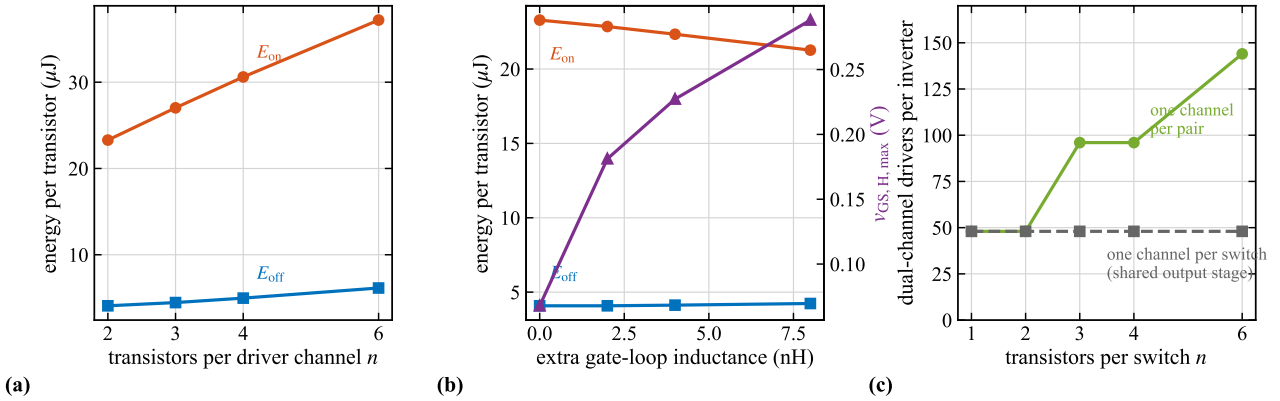


Figure 5: (a) Switching energy per transistor when one driver channel is shared by n transistors (2EDF7275K typical, -1 V, same current per transistor; the output resistance per transistor scales with n). (b) Effect of a longer gate loop (extra inductance in every branch, e.g., a one-driver tree for more devices): switching energy and high-side gate peak. (c) Number of dual-channel drivers of the 48-leg inverter for one channel per pair of transistors and for one channel per switch.

Driving more paralleled transistors from one driver channel fails for four reasons:

1. **Symmetry.** A driver on a mirror line reaches exactly two transistors through identical branches. A third branch cannot be made identical in a planar layout whose gate loops also have to stay orthogonal to the power loops, and any mismatch moves switching energy into one device (Table 1).
2. **Output resistance.** The output stage is shared, so the effective pull-up and pull-down resistance per transistor grows with n . At the same current per transistor, E_{on} rises from 23.3 to 27.0, 30.6 and 37.2 μJ and E_{off} from 4.1 to 6.1 μJ per transistor for $n = 2, 3, 4, 6$ (Fig. 5a): the advantage of paralleling is partly lost in switching loss.
3. **Peak current.** The peak sink current grows to 13.3 A for four transistors at -1 V (Fig. 3c), far above the 8 A rating; limiting it with resistors slows the turn-off further.
4. **Gate-loop length.** A tree that reaches four transistors from one driver is longer. An extra 2, 4 and 8 nH per branch raises the high-side peak from 0.07 to 0.18, 0.23 and 0.29 V, eating into the margin of the negative bias, and reduces the turn-on dv/dt (16.6 to 12.9 V/ns); it also increases the gate ringing and the risk of exceeding the +6 V gate rating at turn-on, which was not evaluated here.

The building block therefore scales in pairs: one driver channel per pair of transistors in a star, three transistors per switch need a second channel, and four are built as two two-transistor structures with two drivers (Fig. 5c). Going from two to three or four transistors per switch doubles the drivers of the inverter from 48 to 96, together with their isolated supplies; with the leg loss falling by only 1.3–1.7 W from two to three or four transistors (design paper), two transistors per switch remain the best choice.

6 Driver comparison

Table 2: Gate drivers considered (values from the datasheets in Datasheet/ where available; n/a: not checked)

	LT8418 (model)	2EDL5013U2D	uP1966E (EPC90156)	2EDF7275K
Type	100 V half bridge, bootstrap	120 V half bridge, bootstrap	80 V half bridge, bootstrap	dual-channel isolated (1.5 kV functional)
Pull-up / pull-down	0.6 / 0.2 Ω (model value)	HS 0.9 / 0.5 Ω , LS 1.5 / 0.5 Ω	n/a	0.85 / 0.35 Ω typ., 1.6 / 0.75 Ω max
Peak source / sink	n/a	2–3 A / 5 A	n/a	4 A / 8 A
Miller clamp	n/a	yes, 0.5 V, 5 ns delay, 7 A	n/a	no
Supply	n/a	4.5–5.5 V (5.9 V abs. max)	5 V gate drive	4.5–20 V, UVLO 4 V
Negative off-voltage	no	no (supply limit)	no	yes, split isolated supply
Use in this work	reference model	bootstrap alternative with-out bias	EPC reference board (1 Ω /0.47 Ω)	selected

7 Recommendations

1. Keep the centre-driver star with Kelvin vias in every source pad; route the Kelvin return directly under the gate trace (1.71 nH, 0.030 nH).
2. Use the 2EDF7275K, one channel per switch star, with a +5/–1 V split isolated supply; keep the dead time short (5–10 ns) to limit the dead-time loss.
3. Limit the peak sink current to the 8 A rating (about 0.4 Ω external turn-off resistor per transistor) or confirm the driver current limit with a current-limited model.
4. Scale paralleling in pairs, one driver channel per pair; do not drive more than two transistors from one channel.
5. On the prototype, measure the high-side gate with an isolated probe at the gate pad, and keep footprints for a gate–source capacitor as a fallback.

Files: simulation/bb_spice/BB_bump.inc with BB_bump.cir, BB_bump_rpd.cir, BB_2EDF7275K.cir, BB_drv_share.cir, BB_gate_len.cir; simulation/sharing/paper_run; simulation/bb_q3d/bb_q3d.cs.py; scripts/gate_driver_report.py.